

$$\begin{array}{ccccccccc}
\pi_{n+1}(Z_{n+1}) & \longrightarrow & \pi_{n+1}(Z_n) & \longrightarrow & \pi_{n+1}(Z_n, Z_{n+1}) & \longrightarrow & \pi_n(Z_{n+1}) & \longrightarrow & \pi_n(Z_n) \\
\downarrow = & & \downarrow \approx & & \downarrow & & \downarrow = & & \downarrow \\
\pi_{n+1}(Z_{n+1}) & \longrightarrow & \pi_{n+1}(Y) & \longrightarrow & \pi_{n+1}(Y, Z_{n+1}) & \longrightarrow & \pi_n(Z_{n+1}) & \longrightarrow & \pi_n(Y) \\
\uparrow & & \uparrow = & & \uparrow & & \uparrow \approx & & \uparrow = \\
\pi_{n+1}(X) & \longrightarrow & \pi_{n+1}(Y) & \longrightarrow & \pi_{n+1}(Y, X) & \longrightarrow & \pi_n(X) & \longrightarrow & \pi_n(Y)
\end{array}$$

The upper-right vertical map is injective and the lower-left vertical map is surjective, so the five-lemma implies that the two middle vertical maps are isomorphisms. Since the homotopy fiber of an inclusion $A \hookrightarrow B$ has π_i equal to $\pi_{i+1}(B, A)$, we see that condition (3) is satisfied.

The statement about a tower of principal fibrations can be obtained as an application of Lemma 4.70. As we saw in the previous paragraph, there are isomorphisms $\pi_{n+1}(Y, X) \approx \pi_{n+1}(Z_n, Z_{n+1})$, and these respect the action of $\pi_1(X) \approx \pi_1(Z_{n+1})$, so Lemma 4.70 gives the result. $\square$

Besides the case that Y is a point, which yields Postnikov towers, another interesting special case of Moore-Postnikov towers is when X is a point. In this case the space Z_n is an n -connected covering of Y , as in Example 4.20. The n -connected covering of Y can also be obtained as the homotopy fiber of the n^{th} stage $Y \rightarrow Y_n$ of a Postnikov tower for Y . The tower of n -connected coverings of Y can be realized by principal fibrations by taking Z_n to be the homotopy fiber of the map $Z_{n-1} \rightarrow K(\pi_n Y, n)$ that is the first nontrivial stage in a Postnikov tower for Z_{n-1} .

$$\begin{array}{c}
\vdots \\
\downarrow \\
Z_2 \rightarrow K(\pi_3 Y, 3) \\
\downarrow \\
Z_1 \rightarrow K(\pi_2 Y, 2) \\
\downarrow \\
Y \rightarrow K(\pi_1 Y, 1)
\end{array}$$

A generalization of the preceding theory allowing nontrivial actions of π_1 can be found in [Robinson 1972].

Obstruction Theory

It is very common in algebraic topology to encounter situations where one would like to extend or lift a given map. Obvious examples are the homotopy extension and homotopy lifting properties. In their simplest forms, extension and lifting questions can often be phrased in one of the following two ways:

The Extension Problem. Given a CW pair (W, A) and a map $A \rightarrow X$, does this extend to a map $W \rightarrow X$?

$$\begin{array}{ccc}
A & \longrightarrow & X \\
\downarrow & \nearrow \text{---} & \\
W & &
\end{array}$$

The Lifting Problem. Given a fibration $X \rightarrow Y$ and a map $W \rightarrow Y$, is there a lift $W \rightarrow X$?

$$\begin{array}{ccc}
& & X \\
& \nearrow \text{---} & \downarrow \\
W & \longrightarrow & Y
\end{array}$$

In order for the lifting problem to include things like the homotopy lifting property, it should be generalized to a relative form:

The Relative Lifting Problem. Given a CW pair (W, A) , a fibration $X \rightarrow Y$, and a map $W \rightarrow Y$, does there exist a lift $W \rightarrow X$ extending a given lift on A ?

$$\begin{array}{ccc} A & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow \\ W & \longrightarrow & Y \end{array}$$

Besides reducing to the absolute lifting problem when $A = \emptyset$, this includes the extension problem by taking Y to be a point. Of course, one could broaden these questions by dropping the requirements that (W, A) be a CW pair and that the map $X \rightarrow Y$ be a fibration. However, these conditions are often satisfied in cases of interest, and they make the task of finding solutions much easier.

The term ‘obstruction theory’ refers to a procedure for defining a sequence of cohomology classes that are the obstructions to finding a solution to the extension, lifting, or relative lifting problem. In the most favorable cases these obstructions lie in cohomology groups that are all zero, so the problem has a solution. But even when the obstructions are nonzero it can be very useful to have the problem expressed in cohomological terms.

There are two ways of developing obstruction theory, which produce essentially the same result in the end. In the more elementary approach one tries to construct the extension or lifting one cell of W at a time, proceeding inductively over skeleta of W . This approach has an appealing directness, but the technical details of working at the level of cochains are perhaps a little tedious. Instead of pursuing this direct line we shall follow the second approach, which is slightly more sophisticated but has the advantage that the theory becomes an almost trivial application of Postnikov towers for the extension problem, or Moore–Postnikov towers for the lifting problem. The cellular viewpoint is explained in [VBKT], where it appears in the study of characteristic classes of vector bundles.

Let us consider the extension problem first, where we wish to extend a map $A \rightarrow X$ to the larger complex W . Suppose that X has a Postnikov tower of principal fibrations. Then we have a commutative diagram as shown below, where we have enlarged the tower by adjoining the space X_0 , which is just a point, at the bottom. The map $X_1 \rightarrow X_0$ is then a fibration, and to say it is principal says that X_1 , which in any case is a $K(\pi_1 X, 1)$, is the loopspace of $K(\pi_1 X, 2)$, hence $\pi_1(X)$ must be abelian. Conversely, if $\pi_1(X)$ is abelian and acts trivially on all the higher homotopy groups of X , then there is an extended Postnikov tower of principal fibrations as shown.

$$\begin{array}{ccccc} & & & & \vdots \\ & & & & \downarrow \\ & & & & X_3 \xrightarrow{k_3} K(\pi_4 X, 5) \\ & & & & \downarrow \\ & & & & X_2 \xrightarrow{k_2} K(\pi_3 X, 4) \\ & & & & \downarrow \\ & & & & X_1 \xrightarrow{k_1} K(\pi_2 X, 3) \\ & & & & \downarrow \\ A \longrightarrow X & \longrightarrow & X_1 & \xrightarrow{k_0} & X_0 \xrightarrow{k_0} K(\pi_1 X, 2) \\ \downarrow & & \downarrow & & \\ W & \longrightarrow & X_0 & & \end{array}$$

Our strategy will be to try to lift the constant map $W \rightarrow X_0$ to maps $W \rightarrow X_n$ for $n = 1, 2, \dots$ in succession, extending the given maps $A \rightarrow X_n$. If we are able to find all these lifts $W \rightarrow X_n$, there will then be no difficulty in constructing the desired extension $W \rightarrow X$.

For the inductive step we have a commutative diagram as at the right. Since X_n is the pullback, its points are pairs consisting of a point in X_{n-1} and a path from its image in K to the basepoint. A lift $W \rightarrow X_n$ therefore amounts to a nullhomotopy of the composition $W \rightarrow X_{n-1} \rightarrow K$. We already have such a lift defined on A , hence a nullhomotopy of $A \rightarrow K$, and we want a nullhomotopy of $W \rightarrow K$ extending this nullhomotopy on A .

$$\begin{array}{ccccc} A & \longrightarrow & X_n & \longrightarrow & PK \\ \downarrow & & \downarrow & & \downarrow \\ W & \longrightarrow & X_{n-1} & \longrightarrow & K = K(\pi_n X, n+1) \end{array}$$

The map $W \rightarrow K$ together with the nullhomotopy on A gives a map $W \cup CA \rightarrow K$, where CA is the cone on A . Since K is a $K(\pi_n X, n+1)$, the map $W \cup CA \rightarrow K$ determines an **obstruction** class $\omega_n \in H^{n+1}(W \cup CA; \pi_n X) \approx H^{n+1}(W, A; \pi_n X)$.

Proposition 4.72. *A lift $W \rightarrow X_n$ extending the given $A \rightarrow X_n$ exists iff $\omega_n = 0$.*

Proof: We need to show that the map $W \cup CA \rightarrow K$ extends to a map $CW \rightarrow K$ iff $\omega_n = 0$, or in other words, iff $W \cup CA \rightarrow K$ is homotopic to a constant map.

Suppose that $g_t: W \cup CA \rightarrow K$ is such a homotopy. The constant map g_1 then extends to the constant map $g_1: CW \rightarrow K$, so by the homotopy extension property for the pair $(CW, W \cup CA)$, applied to the reversed homotopy g_{1-t} , we have a homotopy $g_t: CW \rightarrow K$ extending the previous homotopy $g_t: W \cup CA \rightarrow K$. The map $g_0: CW \rightarrow K$ then extends the given map $W \cup CA \rightarrow K$.

Conversely, if we have an extension $CW \rightarrow K$, then this is nullhomotopic since the cone CW is contractible, and we may restrict such a nullhomotopy to $W \cup CA$. $\square$

If we succeed in extending the lifts $A \rightarrow X_n$ to lifts $W \rightarrow X_n$ for all n , then we obtain a map $W \rightarrow \varprojlim X_n$ extending the given $A \rightarrow X \rightarrow \varprojlim X_n$. Let M be the mapping cylinder of $X \rightarrow \varprojlim X_n$. Since the restriction of $W \rightarrow \varprojlim X_n \subset M$ to A factors through X , this gives a homotopy of this restriction to the map $A \rightarrow X \subset M$. Extend this to a homotopy of $W \rightarrow M$, producing a map $(W, A) \rightarrow (M, X)$. Since the map $X \rightarrow \varprojlim X_n$ is a weak homotopy equivalence, $\pi_i(M, X) = 0$ for all i , so by Lemma 4.6, the compression lemma, the map $(W, A) \rightarrow (M, X)$ can be homotoped to a map $W \rightarrow X$ extending the given $A \rightarrow X$, and we have solved the extension problem.

Thus if it happens that at each stage of the inductive process of constructing lifts $W \rightarrow X_n$ the obstruction $\omega_n \in H^{n+1}(W, A; \pi_n X)$ vanishes, then the extension problem has a solution. In particular, this yields:

Corollary 4.73. *If X is a connected abelian CW complex and (W, A) is a CW pair such that $H^{n+1}(W, A; \pi_n X) = 0$ for all n , then every map $A \rightarrow X$ can be extended to a map $W \rightarrow X$. $\square$*

This is a considerable improvement on the more elementary result that extensions exist if $\pi_n(X) = 0$ for all n such that $W - A$ has cells of dimension $n+1$, which is an exercise for §4.1.

We can apply the Hurewicz theorem and obstruction theory to extend the homology version of Whitehead's theorem to CW complexes with trivial action of π_1 on all homotopy groups:

Proposition 4.74. *If X and Y are connected abelian CW complexes, then a map $f: X \rightarrow Y$ inducing isomorphisms on all homology groups is a homotopy equivalence.*

Proof: Taking the mapping cylinder of f reduces us to the case of an inclusion $X \hookrightarrow Y$ of a subcomplex. If we can show that $\pi_1(X)$ acts trivially on $\pi_n(Y, X)$ for all n , then the relative Hurewicz theorem will imply that $\pi_n(Y, X) = 0$ for all n , so $X \rightarrow Y$ will be a weak homotopy equivalence. The assumptions guarantee that $\pi_1(X) \rightarrow \pi_1(Y)$ is an isomorphism, so we know at least that $\pi_1(Y, X) = 0$.

We can use obstruction theory to extend the identity map $X \rightarrow X$ to a retraction $Y \rightarrow X$. To apply the theory we need $\pi_1(X)$ acting trivially on $\pi_n(X)$, which holds by hypothesis. Since the inclusion $X \hookrightarrow Y$ induces isomorphisms on homology, we have $H_*(Y, X) = 0$, hence $H^{n+1}(Y, X; \pi_n(X)) = 0$ for all n by the universal coefficient theorem. So there are no obstructions, and a retraction $Y \rightarrow X$ exists. This implies that the maps $\pi_n(Y) \rightarrow \pi_n(Y, X)$ are onto, so trivial action of $\pi_1(X)$ on $\pi_n(Y)$ implies trivial action on $\pi_n(Y, X)$ by naturality of the action. $\square$

The generalization of the preceding analysis of the extension problem to the relative lifting problem is straightforward. Assuming the fibration $p: X \rightarrow Y$ in the statement of the lifting problem has a Moore-Postnikov tower of principal fibrations, we have the diagram shown at the right.

The first step is to lift the map $W \rightarrow Y$ to Z_1 , extending the given lift on A . We may take Z_1 to be the covering space of Y corresponding to the subgroup $p_*(\pi_1(X))$ of $\pi_1(Y)$, so covering space theory

$$\begin{array}{ccccc} & & & \vdots & \\ & & & \downarrow & \\ & & & Z_2 & \longrightarrow K(\pi_2 F, 3) \\ & & \nearrow & \downarrow & \\ A & \longrightarrow & X & \longrightarrow & Z_1 & \longrightarrow K(\pi_1 F, 2) \\ \downarrow & & \searrow p & & \downarrow & \\ W & \longrightarrow & & & Y & \end{array}$$

tells us when we can lift $W \rightarrow Y$ to Z_1 , and the unique lifting property for covering spaces can be used to see whether a lift can be chosen to agree with the lift on A given by the diagram; this could only be a problem when A has more than one component.

Having a lift to Z_1 , the analysis proceeds exactly as before. One encounters a sequence of obstructions $\omega_n \in H^{n+1}(W, A; \pi_n F)$, assuming $\pi_1 F$ is abelian in the case $n = 1$. A lift to X exists, extending the given lift on A , if each successive ω_n is zero.

One can ask the converse question: If a lift exists, must the obstructions ω_n all be zero? Since Proposition 4.72 is an if and only if statement, one might expect the answer to be yes, but upon closer inspection the matter becomes less clear. The difficulty is that, even if at some stage the obstruction ω_n is zero, so a lift to Z_{n+1} exists, there may be many choices of such a lift, and different choices could lead to different ω_{n+1} 's, some zero and others nonzero. Examples of such ambiguities are not hard to produce, for both the lifting and the extension problems, and the

ambiguities only become worse with each subsequent choice of a lift. So it is only in rather special circumstances that one can say that there are well-defined obstructions. A simple case is when $\pi_i(F) = 0$ for $i < n$, so the Moore-Postnikov factorization begins with Z_n as in the diagram at the right. In this case the composition across the bottom of the diagram gives a well-defined **primary obstruction** $\omega_n \in H^{n+1}(W, A; \pi_n F)$.

$$\begin{array}{ccccc}
 & & \vdots & & \\
 & & \downarrow & & \\
 & & Z_{n+2} & \longrightarrow & K(\pi_{n+2}F, n+3) \\
 & \nearrow & \downarrow & & \\
 A \longrightarrow X & & Z_{n+1} & \longrightarrow & K(\pi_{n+1}F, n+2) \\
 \downarrow & \searrow p & \downarrow & & \\
 W \longrightarrow & & Y = Z_n & \longrightarrow & K(\pi_n F, n+1)
 \end{array}$$

Exercises

1. Show there is a map $\mathbb{R}P^\infty \rightarrow \mathbb{C}P^\infty = K(\mathbb{Z}, 2)$ which induces the trivial map on $\tilde{H}_*(-; \mathbb{Z})$ but a nontrivial map on $\tilde{H}^*(-; \mathbb{Z})$. How is this consistent with the universal coefficient theorem?
2. Show that the group structure on S^1 coming from multiplication in $\mathbb{C}$ induces a group structure on $\langle X, S^1 \rangle$ such that the bijection $\langle X, S^1 \rangle \rightarrow H^1(X; \mathbb{Z})$ of Theorem 4.57 is an isomorphism.
3. Suppose that a CW complex X contains a subcomplex S^1 such that the inclusion $S^1 \hookrightarrow X$ induces an injection $H_1(S^1; \mathbb{Z}) \rightarrow H_1(X; \mathbb{Z})$ with image a direct summand of $H_1(X; \mathbb{Z})$. Show that S^1 is a retract of X .
4. Given abelian groups G and H and CW complexes $K(G, n)$ and $K(H, n)$, show that the map $\langle K(G, n), K(H, n) \rangle \rightarrow \text{Hom}(G, H)$ sending a homotopy class $[f]$ to the induced homomorphism $f_* : \pi_n(K(G, n)) \rightarrow \pi_n(K(H, n))$ is a bijection.
5. Show that $[X, S^n] \approx H^n(X; \mathbb{Z})$ if X is an n -dimensional CW complex. [Build a $K(\mathbb{Z}, n)$ from S^n by attaching cells of dimension $\geq n+2$.]
6. For an abelian group G , Theorem 4.57 gives a map $\mu : K(G, n) \times K(G, n) \rightarrow K(G, n)$ with $\mu^*(\alpha) = \alpha \times 1 + 1 \times \alpha$ where α is a fundamental class for $K(G, n)$. Show that μ defines an H-space structure on $K(G, n)$ that is commutative and associative, up to homotopy. Show also that the H-space multiplication μ is unique up to homotopy.
7. Using an H-space multiplication μ on $K(G, n)$, define an addition in $\langle X, K(G, n) \rangle$ by $[f] + [g] = [\mu(f, g)]$ and show that under the bijection $H^n(X; G) \approx \langle X, K(G, n) \rangle$ this addition corresponds to the usual addition in cohomology.
8. Show that a map $p : E \rightarrow B$ is a fibration iff the map $\pi : E^I \rightarrow E_p$, $\pi(y) = (y(0), py)$, has a section, that is, a map $s : E_p \rightarrow E^I$ such that $ps = 1$.
9. Show that a linear projection of a 2-simplex onto one of its edges is a fibration but not a fiber bundle. [Use the preceding problem.]
10. Given a fibration $F \rightarrow E \rightarrow B$, use the homotopy lifting property to define an action of $\pi_1(E)$ on $\pi_n(F)$, a homomorphism $\pi_1(E) \rightarrow \text{Aut}(\pi_n(F))$, such that the composition $\pi_1(F) \rightarrow \pi_1(E) \rightarrow \text{Aut}(\pi_n(F))$ is the usual action of $\pi_1(F)$ on $\pi_n(F)$. Deduce that if $\pi_1(E) = 0$, then the action of $\pi_1(F)$ on $\pi_n(F)$ is trivial.